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$$w = \ln(x^2 + y^2 + z^2) \text{ met } x = s - t, y = s + t \text{ en } z = 2\sqrt{st}$$

$$\frac{\partial w}{\partial x} = \frac{2x}{x^2 + y^2 + z^2}$$

$$\frac{\partial w}{\partial y} = \frac{2y}{x^2 + y^2 + z^2}$$

$$\frac{\partial w}{\partial z} = \frac{2z}{x^2 + y^2 + z^2}$$

Eerst voor s :

$$\frac{\partial x}{\partial s} = 1$$

$$\frac{\partial y}{\partial s} = 1$$

$$\frac{\partial z}{\partial s} = \frac{t}{\sqrt{st}}$$

$$Dw(x, y, z) = \left(\frac{2x}{x^2 + y^2 + z^2} \quad \frac{2y}{x^2 + y^2 + z^2} \quad \frac{2z}{x^2 + y^2 + z^2} \right)$$

$$D(x, y, z)(s) = \begin{pmatrix} 1 \\ 1 \\ \frac{t}{\sqrt{st}} \end{pmatrix}$$

$$Dw(x, y, z) \cdot D(x, y, z)(s) = \frac{2x + 2y + \frac{2zt}{\sqrt{st}}}{x^2 + y^2 + z^2}$$

$$= \frac{2(s - t) + 2(s + t) + 4t}{(s - t)^2 + (s + t)^2 + 4st} = \frac{2}{(s + t)}$$

$$\frac{\partial w}{\partial s} = \frac{2}{(s + t)}$$

Nu voor t :

$$\frac{\partial x}{\partial t} = -1$$

$$\frac{\partial y}{\partial t} = 1$$

$$\frac{\partial z}{\partial t} = \frac{s}{\sqrt{st}}$$

$$D(x, y, z)(t) = \begin{pmatrix} -1 \\ 1 \\ \frac{s}{\sqrt{st}} \end{pmatrix}$$

$$\begin{aligned}
Dw(x, y, z) \cdot D(x, y, z)(t) &= \frac{-2x + 2y + \frac{2zs}{\sqrt{st}}}{x^2 + y^2 + z^2} \\
&= \frac{-2(s-t) + 2(s+t) + 4s}{2(s+t)^2} = \frac{2}{s+t} \\
\frac{\partial w}{\partial t} &= \frac{2}{s+t}
\end{aligned}$$

References